

RISING BEYOND BORDERS

Document 17 — Analytics, Privacy Technology & Production Observability
Master Specification + Claude Code Implementation Prompt

Status: Implementation specification

Purpose: Define a safe production architecture for analytics, privacy-related technology, search visibility, error monitoring and performance observability. This phase does not activate tracking or select vendors on RBB's behalf.

1. Phase Objective

Documents 13–16 established policy dependencies, production rendering and accessibility. This phase separates **what the website technically can measure** from **what RBB has approved to collect**. Until those decisions are made, the site should remain minimally instrumented and should not silently add analytics or tracking.

2. Current Baseline

- Previous implementation reported no analytics, maps, video embeds or browser storage in the application.
- Document 14 deliberately left analytics and Search Console unconfigured.
- Document 13 identified third-party font requests and an unresolved hosting technology audit.
- Document 15 established that hosting-generated cookies/scripts must be checked after the host is selected.
- Forms remain disabled until RBB approves a submission destination and privacy/consent requirements.

3. Core Principle: No Unapproved Tracking

- Do not add Google Analytics, Meta Pixel, advertising pixels, heatmaps, session replay or similar tracking without explicit RBB approval.
- Do not create hidden tracking through a dependency merely because it is convenient.
- Do not place analytics identifiers in source code unless the service and identifier have been approved.
- Do not treat Search Console verification as equivalent to visitor analytics.
- Do not activate optional tracking merely because the production host supports it.
- Until approved, public pages should work without optional analytics cookies or browser-storage tracking.

4. Technology Categories

Category	Default state	Decision needed
Essential hosting/runtime	Allowed only as required for site delivery	Host audit
Analytics	Off	RBB chooses provider/purpose
Search Console	Unconfigured	RBB approves ownership/verification
Performance monitoring	Off until approved	RBB chooses scope/provider
Error monitoring	Off until approved	RBB chooses provider/data scope
Session replay/heatmaps	Off	Explicit approval required
Advertising/remarketing	Off	Explicit approval required
Fonts/CDN	Existing implementation; audit required	Host/font/privacy review

Forms observability	Not active	Depends on form backend
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5. Analytics Architecture

If RBB later approves analytics, implementation should use a single abstraction layer rather than scattering vendor calls throughout React components.

- Create a small analytics interface for page views and explicitly approved events.
- Components call the internal interface; the provider adapter owns vendor-specific code.
- Analytics must be disabled by configuration when not approved.
- Do not put personal form contents, donation details, message text or sensitive fields into analytics events.
- Do not use URL query parameters to leak form values into analytics.
- Use route/page identifiers that describe public content rather than personal data.
- Document every event before implementation.

6. Event Governance

- Each event needs: name, purpose, trigger, properties, data classification and retention requirement.
- Keep the initial event vocabulary small.
- Recommended initial candidates, only if approved: `page_view`, `navigation_interaction`, `outbound_link`, `content_select` and approved CTA interaction.
- Do not automatically capture every click, keystroke, scroll position or form field.
- Do not capture free-text form values.
- Do not send email addresses, phone numbers, names, addresses or message bodies to analytics.
- Do not create events for hidden development/catalogue content.

7. Consent & Privacy Dependency

The public privacy policy remains pending. Therefore this document defines architecture, not a legal conclusion about consent requirements. Before optional tracking is activated, RBB should obtain appropriate legal/privacy review for its actual audience, jurisdiction, provider contracts and technology.

- Separate essential technology from optional measurement.
- Optional tracking must have an approved consent approach where required.
- Consent state must be respected before optional trackers initialize.
- Do not load optional analytics first and ask for consent afterward.
- Do not store consent decisions in an undisclosed mechanism.
- If a consent-management platform is selected, document its own cookies, scripts and data flows.

8. Cookie & Browser Storage Audit

- After the hosting provider is selected, inspect response headers and browser storage on a clean session.
- Record cookies by name, purpose, domain, duration and whether they are essential or optional.
- Inspect `localStorage`, `sessionStorage`, `IndexedDB` and service-worker behavior.
- Inspect third-party requests and identify their owner/purpose.

- Do not publish a cookie statement based only on source-code inspection.
- Repeat the audit after analytics, forms or other third-party services are introduced.

9. Font & Third-Party Resource Audit

- Document 15 identified external font traffic and a possible future self-hosting decision.
- Once hosting is selected, record all third-party font/resource requests.
- Confirm whether Satoshi may be self-hosted before making that change.
- If third-party resources remain, document their purpose and dependency in the technology audit.
- Do not add unrelated third-party libraries for convenience.

10. Google Search Console / Search Visibility

- Search Console is an external site-management/diagnostic service, not visitor analytics.
- Do not add Search Console verification until the production domain is confirmed.
- Choose one approved verification method compatible with the selected host.
- After verification, inspect indexing, sitemap processing and coverage.
- Do not put Search Console identifiers into public analytics events.
- Search Console does not replace the site's own sitemap or metadata implementation.

11. Error Monitoring

- Error monitoring is optional and must be explicitly approved.
- If selected, capture technical error context without form contents or other unnecessary personal data.
- Do not send full request bodies, authorization headers, cookies or secrets.
- Prefer normalized error categories and route information.
- Source maps, if uploaded to a monitoring vendor, must be protected according to the vendor's access controls.
- Do not expose monitoring DSNs or configuration as secrets unless the vendor explicitly requires them; distinguish public identifiers from secret credentials.

12. Performance Observability

- Document 15 already defines $LCP \leq 2.5s$, $INP \leq 200ms$ and $CLS \leq 0.1$ as engineering targets.
- Production performance should be measured again on the real deployed site.
- Prefer privacy-conscious aggregate measurements.
- Do not introduce session replay simply to understand performance.
- Record test environment and population before interpreting performance data.
- Use lab measurements for release QA and field measurements only when RBB approves the collection technology.

13. Uptime & Availability Monitoring

- An external uptime checker may be useful after production deployment, but RBB must approve the provider.
- Use a public health endpoint or normal public page rather than a form endpoint that could create submissions.
- Do not expose secrets through health endpoints.

- Document check frequency, region(s), alert recipients and retention if a provider is selected.
- Uptime monitoring does not require visitor analytics.

14. Form Observability

- Forms remain disabled until a backend/submission service is approved.
- Once active, monitor operational failures without logging submitted message bodies.
- Never send raw form fields to analytics.
- Application logs should redact or avoid personal data where possible.
- Rate-limit and anti-spam outcomes may be logged as aggregate technical events if approved.
- Do not log authentication tokens, cookies or provider secrets.

15. Donation / Payment Observability

- Donation/payment tracking must not be designed until the actual payment provider and donation flow are approved.
- Do not send card/payment details to analytics or application logs.
- Do not infer donation amounts from client-side events as a source of financial truth.
- Payment provider webhooks, if later introduced, should be handled server-side and logged minimally.
- Analytics should distinguish an intent/CTA interaction from a verified successful donation.

16. Environment Separation

- Development must not send production analytics by default.
- Preview/staging must use separate identifiers or remain disabled.
- Production identifiers belong only in production configuration.
- Never commit secrets or private provider credentials.
- Build output should make it possible to verify whether optional tracking is enabled.
- Do not use a production domain or analytics ID as a development default.

17. Privacy-Safe Logging

- Do not log passwords, access tokens, authorization headers, payment details or free-text form submissions.
- Minimize URLs when query strings may contain personal information.
- Redact email addresses, phone numbers and similar identifiers when technical logs do not require them.
- Keep logs only as long as operationally necessary under the chosen provider's retention controls.
- Document who can access production logs.

18. Observability Data Model

- Separate product analytics, application errors, performance metrics and uptime checks.
- Each stream should have its own purpose and retention policy.
- Do not merge all telemetry into one unrestricted data collection system.
- Use stable route/content IDs rather than personal identifiers.
- Record provider, region, retention, access and deletion capabilities when a vendor is selected.

19. Security Boundaries

- Analytics is not an authorization system.
- Client-side event data must never be treated as trusted business data.
- Never expose provider API secrets in the browser.
- Do not use analytics to transmit security-sensitive information.
- Monitoring endpoints must not become a way to expose internal configuration.
- Keep third-party scripts to the minimum approved set.

20. Production Technology Audit

- After the host is selected, crawl the deployed site with a clean browser profile.
- Record cookies and storage.
- Record all third-party network requests.
- Record script and font origins.
- Record analytics/monitoring IDs and whether they load before or after consent where applicable.
- Compare actual behavior against the approved privacy policy and technology inventory.
- Repeat after any major third-party integration.

21. RBB Decision Matrix

Decision	Current state
Production domain	Pending
Hosting provider	Pending
Analytics provider	Pending
Analytics purposes/events	Pending
Consent approach	Pending legal/privacy review
Search Console verification	Pending domain/host
Error monitoring	Pending
Performance/field monitoring	Pending
Uptime monitoring	Pending
Form provider	Pending
Donation/payment provider	Pending
Satoshi self-hosting	Pending licence confirmation
Accessibility statement	Pending

22. Claude Code Implementation Prompt

You are implementing Document 17 – Analytics, Privacy Technology & Production Observability for the Rising Beyond Borders website.

CONTEXT

- Documents 01-16 are implemented and approved.
- Analytics, Search Console, monitoring and production hosting are NOT yet approved.
- Do not activate tracking.
- Do not choose vendors.
- Do not invent legal/privacy conclusions.
- Do not stage or commit changes.

GOAL

Prepare a vendor-neutral, privacy-conscious observability architecture without enabling optional tracking.

CURRENT STATE

- No analytics should be active.
- No advertising pixels/session replay/heatmaps.
- Forms remain disabled.
- Production domain is not configured.
- Hosting is not selected.
- Privacy/terms/accessibility public pages remain pending where specified.

IMPLEMENT

1. Create a small internal analytics abstraction only if useful, with a disabled/no-op default.
2. Provider-specific code must be isolated behind the abstraction.
3. Do not add a provider SDK just to create the abstraction.
4. Ensure development and preview cannot accidentally send production analytics.
5. Ensure no personal form data enters analytics or logs.
6. Do not capture keystrokes, free-text fields, emails, phone numbers, addresses, payment data or message bodies.
7. Keep Search Console unconfigured until domain approval.
8. Keep error/performance/uptime monitoring unconfigured unless explicitly approved.
9. Do not add cookies or browser storage for optional tracking.
10. Preserve existing rendering, SEO, accessibility and performance behavior.

AUDIT SUPPORT

Add documentation/configuration hooks that make it easy to record:

- third-party scripts
- fonts
- cookies
- localStorage/sessionStorage/IndexedDB
- analytics IDs
- monitoring IDs
- provider origins
- retention/access assumptions

ENVIRONMENT

- Production analytics must default OFF.
- No provider secrets in VITE_* variables.
- Preview/development must not use production IDs.
- Build output should make enabled/disabled state verifiable.

LOGGING

- No passwords/tokens/payment details.
- No raw form submissions.
- No authorization headers/cookies.
- Minimize query-string logging.
- Prefer route IDs and technical error categories.

PERFORMANCE

- Preserve Document 15 targets:
LCP <= 2.5s
INP <= 200ms
CLS <= 0.1
- Do not add session replay.
- Do not add heavy monitoring scripts.

VALIDATION

1. Production build passes with no warnings.
2. Confirm no analytics SDK loads.
3. Confirm no optional tracking requests occur.
4. Confirm no new cookies/storage are created by the application.
5. Confirm no personal data is emitted into console/logging.
6. Confirm existing pre-rendered HTML remains intact.
7. Confirm SEO metadata/sitemap behavior remains intact.
8. Confirm accessibility tests remain clean.
9. Confirm all routes and redirects remain intact.
10. Confirm forms remain disabled.

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11. Inspect bundle for accidental analytics/monitoring dependencies.
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REPORT
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Return:
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- files created/modified  
- architecture chosen  
- proof that tracking remains disabled  
- dependency changes  
- storage/cookie findings  
- network-request findings  
- validation results  
- decisions still requiring RBB approval
```

```
Do not stage or commit.
```

23. Acceptance Checklist

- ✓ No optional analytics activated.
- ✓ No advertising/session-replay/heatmap tooling activated.
- ✓ Search Console remains pending until domain approval.
- ✓ Forms remain disabled.
- ✓ No personal/free-text form data enters analytics or logs.
- ✓ No secrets are exposed to client code.
- ✓ Development/preview cannot silently use production identifiers.
- ✓ No new optional cookies/storage are introduced.
- ✓ Third-party resources can be inventoried.
- ✓ Production performance can be measured later without redesigning the app.
- ✓ Existing SEO, pre-rendering and accessibility behavior remains intact.
- ✓ Vendor selection and legal/privacy decisions remain with RBB.

Important: This phase is intentionally conservative. It prepares the application for responsible production observability but does not authorize RBB's collection of visitor data. Vendor, purpose, consent, retention and public-policy decisions must be approved before optional tracking is enabled.